CYBER MIND SPACE
Module 20 - Cryptography

Terms -> module 1 -> module 20

- Clear text / plaintext: the unencrypted data
- Cipher text: the encrypted data
- Key: specifies the transformation of data for encryption / decryption ("key" is not synonymous with "password", although a password can in fact be used as a key)
- Cipher: an algorithmforperforming encryption and decryption

Symmetric cryptography

- Use the same key for the encryption and the decryption
- Symmetric-key either use stream cipher and block cipher
- Popularalgorithms:AES, DES

Asymmetric / Public Key cryptography

- Two key used: public and private
- Public key is publicly known to everyone, issued by Public Key Infrastructure (PKI) and use to encrypt the data
- Private key is a secret for the public,only known by the owner and it is used to decrypt the data
- Asymmetric cryptography delivers confidentiality, integrity, authenticity and non-repudiation
- Popular algorithms : RSA, DSA and Diffie-Hellman, ECDHA

Substitution Cipher

- Every character is substituted with another one
- More on *Wikipedia*
- Example cipher : *Caesar cipher*

Example:

```
Plaintext : THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG
Ciphertext : QEB NRFZH YOLTK CLU GRJMP LSBO QEB IXWV ALD

Key : right shift of 3
```

Transposition Cipher

- The positions held by units of plaintext are shifted according to a regular system
- Example cipher *Rail Fence cipher*

Example:

```
Clear text: WE ARE DISCOVERED. FLEE AT ONCE

W . . . E . . . C . . . R . . . L . . . T . . .          00.........00.........00
E                                                        ...00....00....00....00...
. E . R . D . S . O . E . E . F . E . A . O . C           ......00.........00......
.
Ciphertext: WECRLTEERDSOEEFEAOCAIVDEN . . . N .
.
```

Polyalphabetic Cipher

- Based on substitution
- Using multiple substitution alphabets
- Examplecipher : *Vigenère cipher*

Stream Cipher

- Text digits are combined with a pseudorandom cipher digit stream (keystream)
- Each plaintext digit is encrypted one at a time with the corresponding digit of the stream
- Examplecipher:RC4, Salsalsa 20, Cacha20

Block Cipher

- Operating on fixed-length groups of bits, called a block, with an unvarying transformation that is specified by a symmetric key
- Example cipher:AES,DES, 3DES, 2DES

Symmetric Algorithms

Data Encryption Standard (DES)

- Introduced in 1975
- Standardized in 1977 by NIST
- Problem with DES: short key length (56 bits) -> ASICS Chips

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- Now considered as insecure
- Improved version: Triple DES (involves DES three times)
- Problem withTriple DES: slow, compute heavy

Parameters

Parameter	Value
Block size	64 bits
Key size	56 bits
No. of rounds	16

Advanced Encryption Standard (AES)

- First published in 1998-1999 - 2000
- Became a federal government standard in 2002
- Firstapproved(and only) publicly accessible cipher approved by the NSA for top secret information

Parameters

Parameter	AES-128 value	AES-192 value	AES-256 value
Block size	128 bits	128 bits	128 bits
Key size	128 bits	192 bits	256 bits
No. of rounds	10	12	14

Modes of Operations

Electronic Code Book (ECB)
Cipher Block Chaining (CBC)
Output Feedback Mode (OFB)
Galois/Counter Mode (GCM)

Hashing

Generating a unique Alphanumeric String for a short of Characters, Program, Application, Files, etc. Avalanche Effect --> If you change a binary bit, Hash Value Will Change Drastically. This is Called Avalanche Effect. Collision →

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